

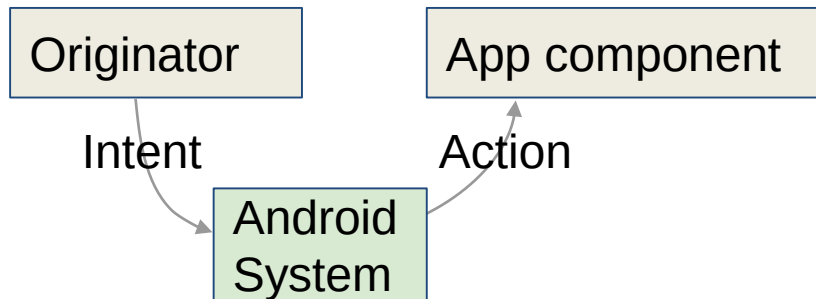
Intent & Connecting Activities

Contents

- **Intent**
- **Starting a new Activity with an Intent**
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- **Navigating between activities**

Intent

- An Intent is a description of an operation to be performed.
- An Intent is an object used to request an action from another app component via the Android system.



What can intents do?

- **Start an Activity**

- A button click starts a new **Activity** for text entry
- Clicking Share opens an app that allows you to post a photo

- **Start a Service**

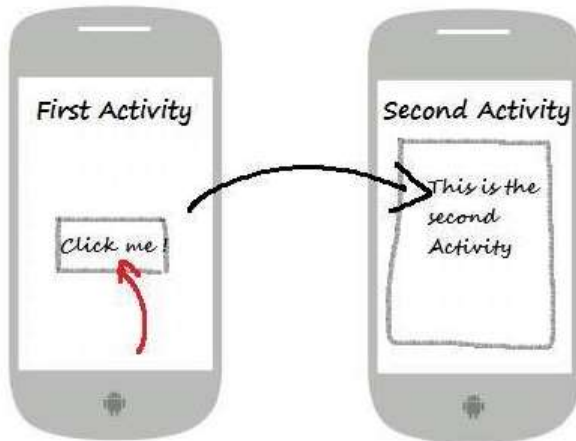
- Initiate downloading a file in the background

- **Deliver Broadcast**

- The system informs everybody that the phone is now charging

INTENT

- **Explicit Intent:** Starts a specific Activity
- **Implicit Intent:** Asks system to find an Activity that can handle this request



INTENT

- **Explicit Intent**

```
Intent i = new Intent(FirstActivity.this, SecondActivity.class);  
startActivity(i);
```

- **Implicit Intent**

```
Intent read1=new Intent();  
read1.setAction(android.content.Intent.ACTION_VIEW);  
read1.setData(ContactsContract.Contacts.CONTENT_URI);  
startActivity(read1);
```

Implicit Intents - Examples

- **Show a web page**

```
Uri uri = Uri.parse("http://www.google.com");  
Intent it = new Intent(Intent.ACTION_VIEW,uri);  
startActivity(it);
```

- **Dial a phone number**

```
Uri uri = Uri.parse("tel:8005551234");  
Intent it = new Intent(Intent.ACTION_DIAL, uri);  
startActivity(it);
```

Activity Implement

- **Start another Activity**

```
Intent intent = new Intent(LoginActivity.this, ProfileActivity.class);  
startActivity(intent);
```

- LoginActivity.this: Current activity
ProfileActivity.class: The activity need to launch

Sending and retrieving data

- **In the first (sending) Activity:**
 - Create the Intent object
 - Put data or extras into that Intent
 - Start the new Activity with `startActivity()`
- **In the second (receiving) Activity:**
 - Get the Intent object, the Activity was started with
 - Retrieve the data or extras from the Intent object

Communicating Between Activities

- Pass data from Activity A to Activity B.

```
Intent intent = new Intent(LoginActivity.this, ProfileActivity.class);  
intent.putExtra("Key", "Value");  
startActivity(intent);
```

- Get data from Activity A at Activity B.

```
if (getIntent() != null) {  
    String data = getIntent().getStringExtra("Key");  
}
```

Communicating Between Activities

- **Pass data from Activity A to Activity B and Receive a result back, using:**

```
public void startActivityForResult (Intent intent, int requestCode)
public void startActivityForResult (Intent intent, int requestCode, Bundle options)
```

- Activity A starts activity B

```
Intent intent=new Intent(MainActivity.this,SecondActivity.class);
startActivityForResult(intent, 2);// Activity is started with requestCode 2
```

- Activity B sends result

```
Intent intent=new Intent();
intent.putExtra("MESSAGE",message);
setResult(2,intent);
finish();//finishing activity
```

- Activity A receives the result back

Communicating Between Activities

- **Pass data from Activity A to Activity B and Receive a result back (cont'd)**
 - Activity A receives the result back

```
@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data)
{
    super.onActivityResult(requestCode, resultCode, data);
    // check if the request code is same as what is passed here it is 2
    if(requestCode==2)
    {
        String message=data.getStringExtra("MESSAGE");
        textView1.setText(message);
    }
}
```

Communicating Between Activities startAcvityForResult is deprecated

```
private ActivityResultLauncher<Intent> activityResultLauncher;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    // Register the activity result launcher
    activityResultLauncher = registerForActivityResult(new ActivityResultContracts.StartActivityForResult(),
        new ActivityResultCallback<ActivityResult>() {
            @Override
            public void onActivityResult(ActivityResult result) {
                if (result.getResultCode() == Activity.RESULT_OK) {
                    Intent data = result.getData();
                    // Handle the returned data
                }
            }
        }
    );
};
```

Communicating Between Activities `startActivityForResult` is deprecated

// Example: Launch another activity

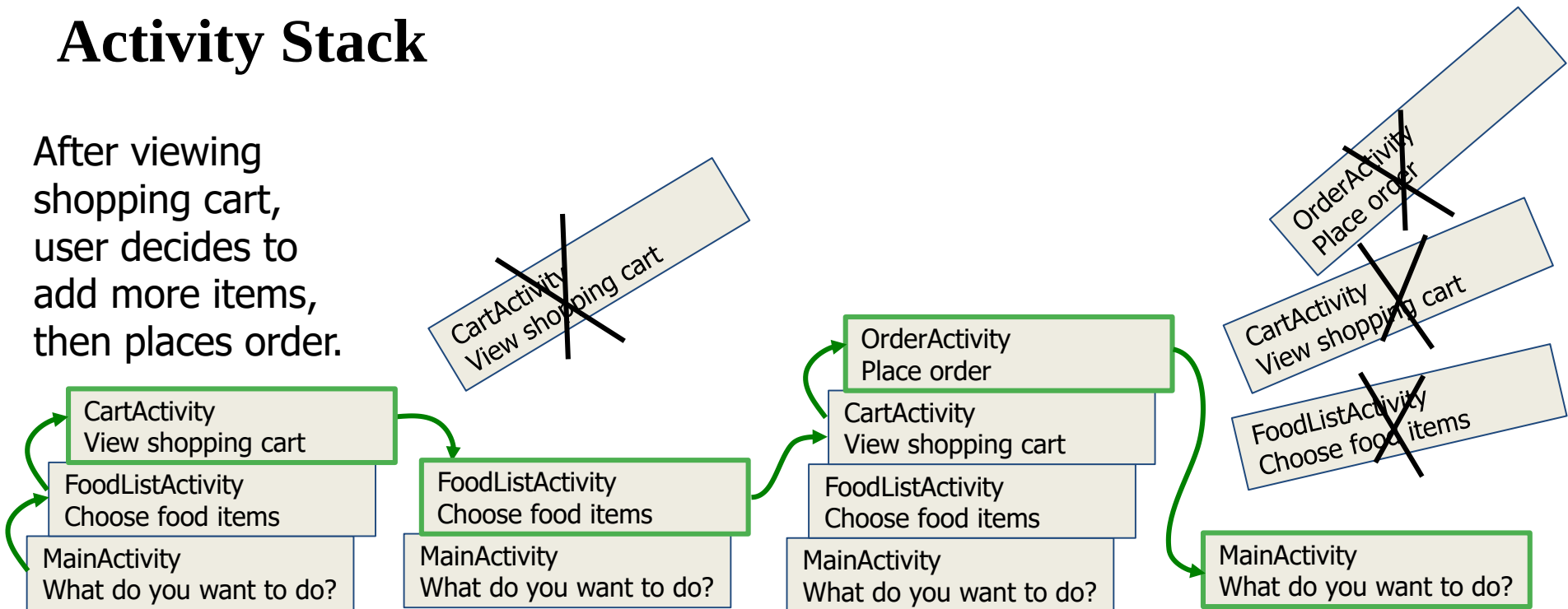
```
findViewById(R.id.btn_go).setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        Intent intent = new Intent(MainActivity2.this, MainActivity.class);  
        activityResultLauncher.launch(intent);  
    }  
});
```

Activity stack

- When a new Activity is started, the previous Activity is stopped and pushed on the Activity back stack
- Last-in-first-out-stack—when the current Activity ends, or the user presses the Back button, it is popped from the stack and the previous Activity resumes

Activity Stack

After viewing shopping cart, user decides to add more items, then places order.



Classwork

- Create LoginActivity:
 - Enter name and touch Login Button to close Login Activity & open HomeActivity
 - Send name to HomeActivity
 - Clear field name if minimize and re-open app
- Create HomeActivity:
 - Get name from LoginActivity and show it on screen
 - Press CLOSE button to finish app

References

- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>